Supplementary Material A for *On the Minimum Distance of High-rate Moore-Type Quasi-Cyclic LDPC Codes*

Complete Proof of Proposition 29: Classification of Weight-8 Matching Configurations

1 Scope and notation

This document gives the complete enumeration proof of Proposition 29.
Its assumptions, notation, and case labels follow the main manuscript.

Let m be an odd prime and let $a, b \in \mathbb{Z}_m^*$ have orders k and j, respectively, where $3 \le j < k$. A column of Moore$(m, a, b, 3)$ is represented by

$$c(x,t) = \left[x,\ x + (b-1)a^t,\ x + (b^2-1)a^t\right]^T, \qquad x \in \mathbb{Z}_m, \quad t \in \mathbb{Z}_k.$$

For two support columns $c_i = c(x_i, t_i)$ and $c_q = c(x_q, t_q)$ and a level $s \in \{0, 1, 2\}$, we write $c_i \sim_s c_q$ if

$$x_i + (b^s - 1)a^{t_i} \equiv x_q + (b^s - 1)a^{t_q} \pmod{m}.$$

For each level s, the equivalence classes of $\sim_s$ form the matching partition M_s, and $M = (M_0, M_1, M_2)$ is the matching configuration.

The proof below uses the following facts established in the main manuscript.

(i) **Column uniqueness (Lemma 13).** Two represented columns satisfy $c(x_1, t_1) = c(x_2, t_2)$ if and only if $x_1 = x_2$ and $t_1 = t_2$.

(ii) **Repeated-pair criterion (Lemma 14).** If two distinct support columns match at two distinct levels, then they are equal; hence such a candidate support is impossible.

(iii) **Isomorphism reduction (Lemma 19).** Weight-preserving isomorphic matching configurations describe the same support configurations up to a relabeling of the support columns.

(iv) **Weight-8 block-size restriction (Lemma 20).** At each of the three matching levels, the block-size type is either $2+2+2+2$ or $4+2+2$.

(v) **Normalization (Corollary 23).** Up to an affine permutation preserving the codeword-support property and Hamming weight, one may assume $c_1 = c(0, 0)$.

(vi) **Completeness of the enumeration (Algorithm 1).** Every even matching block admits a refinement into disjoint 2-matchings. Enumerating all pairwise refinements, all same-level mergings, closing under forced matchings, discarding repeated-column branches, and then reducing by weight-preserving isomorphism yields all legal matching configurations.

2 Statement of Proposition 29

Proposition 1 (Proposition 29 of the main manuscript)**.** *Let* $\boldsymbol{c}$ *be a codeword of* $\mathcal{M}oore(m, a, b, 3)$ *with Hamming weight* 8*. Then, without loss of generality, the matching configuration of* $\boldsymbol{c}$ *is one of the following seven forms.*

Case I:

$$\begin{cases} M_0 = \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 = \{\{c_1, c_3\}, \{c_2, c_4\}, \{c_5, c_7\}, \{c_6, c_8\}\}, \\ M_2 = \{\{c_1, c_4, c_6, c_7\}, \{c_2, c_8\}, \{c_3, c_5\}\}. \end{cases}$$

Case II:

$$\begin{cases} M_0 = \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 = \{\{c_1, c_3, c_6, c_8\}, \{c_2, c_5\}, \{c_4, c_7\}\}, \\ M_2 = \{\{c_1, c_4\}, \{c_2, c_3\}, \{c_5, c_8\}, \{c_6, c_7\}\}. \end{cases}$$

Case III:

$$\begin{cases} M_0 = \{\{c_1, c_2\}, \{c_3, c_4, c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 = \{\{c_1, c_3\}, \{c_2, c_5\}, \{c_4, c_7\}, \{c_6, c_8\}\}, \\ M_2 = \{\{c_1, c_4\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_5, c_8\}\}. \end{cases}$$

Case IV:

$$\begin{cases} M_0 = \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 = \{\{c_1, c_3\}, \{c_2, c_4\}, \{c_5, c_7\}, \{c_6, c_8\}\}, \\ M_2 = \{\{c_1, c_5\}, \{c_2, c_7\}, \{c_3, c_6\}, \{c_4, c_8\}\}. \end{cases}$$

Case V:

$$\begin{cases} M_0 = \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 = \{\{c_1, c_3\}, \{c_2, c_4\}, \{c_5, c_7\}, \{c_6, c_8\}\}, \\ M_2 = \{\{c_1, c_5\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_4, c_8\}\}. \end{cases}$$

Case VI:

$$\begin{cases} M_0 = \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 = \{\{c_1, c_3\}, \{c_2, c_6\}, \{c_5, c_7\}, \{c_4, c_8\}\}, \\ M_2 = \{\{c_1, c_5\}, \{c_2, c_4\}, \{c_3, c_7\}, \{c_6, c_8\}\}. \end{cases}$$

Case VII:

$$\begin{cases} M_0 = \{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\}, \\ M_1 = \{\{c_1, c_3\}, \{c_2, c_7\}, \{c_4, c_5\}, \{c_6, c_8\}\}, \\ M_2 = \{\{c_1, c_5\}, \{c_2, c_6\}, \{c_3, c_7\}, \{c_4, c_8\}\}. \end{cases}$$

3 Complete enumeration proof

Proof. Let $\boldsymbol{c}$ be a codeword of $\mathcal{M}oore(m, a, b, 3)$ of Hamming weight 8. By Corollary 23 of the main manuscript, we may assume without loss of generality that $c_1 = c(0, 0)$.

By Lemma 20 of the main manuscript, at each matching level only the block-size types $2+2+2+2$ and $4+2+2$ can occur. By Algorithm 1 of the main manuscript, it is sufficient to begin with pairwise refinements; the possible and forced mergings are treated below.

Enumeration procedure. Every even matching block admits a pairwise refinement. We enumerate these refinements, include same-level mergings, and close each candidate under the matchings forced by its congruences, as in Algorithm 1 of the main manuscript. At each stage,

repeated-column tests discard impossible branches, and relabelings preserving the fixed levels identify equivalent choices. The calculations below give the surviving configurations; the final isomorphism reduction identifies seven representatives.

After relabeling the support columns, we may refine the matching at level $s = 0$ as

$$M_0^{\mathrm{f}} = \big\{\{c_1, c_2\}, \{c_3, c_4\}, \{c_5, c_6\}, \{c_7, c_8\}\big\}.$$

Equivalently, the first row of the support matrix is written as

$$0, 0, x, x, y, y, z, z.$$

Here M_0^{f} denotes the pairwise refinement used in Algorithm 1; the actual matching partition remains M_0. We do not assume that $0, x, y, z$ are pairwise distinct. Thus the support matrix has the form

$$\mathrm{supp}(\mathbf{c}) = [c_1, c_2, \ldots, c_8],$$

where

$$\mathrm{supp}(\mathbf{c}) = \begin{bmatrix} 0 & 0 & x & x \\ b-1 & (b-1)a^{t_2} & x+(b-1)a^{t_3} & x+(b-1)a^{t_4} \\ b^2-1 & (b^2-1)a^{t_2} & x+(b^2-1)a^{t_3} & x+(b^2-1)a^{t_4} \\ y & y & z & z \\ y+(b-1)a^{t_5} & y+(b-1)a^{t_6} & z+(b-1)a^{t_7} & z+(b-1)a^{t_8} \\ y+(b^2-1)a^{t_5} & y+(b^2-1)a^{t_6} & z+(b^2-1)a^{t_7} & z+(b^2-1)a^{t_8} \end{bmatrix}.$$

By Lemma 14 of the main manuscript, if the same pair of distinct support columns is matched at two different levels, then the two columns coincide. Therefore, every branch containing such a repeated pair is discarded.

At each step below, we enumerate all admissible pairwise matchings which contain no pair already matched at a previous level. The residual relabeling symmetries preserving the previously fixed matchings are used to identify equivalent choices.

Since the labels of the support columns are arbitrary, we may assume that c_1 is matched with c_3 at level $s = 1$. At level $s = 2$, the column c_1 cannot be matched with c_2 or c_3. Up to the remaining relabeling symmetries, there are only two possibilities: either c_1 is matched with the other column in the pair containing c_3, or with a column in another pair.
Hence we obtain the following two principal cases:

$$\text{Case 1} \begin{cases} b-1 & \equiv x+(b-1)a^{t_3} \pmod m, \\ b^2-1 & \equiv x+(b^2-1)a^{t_4} \pmod m, \end{cases} \quad \text{Case 2} \begin{cases} b-1 & \equiv x+(b-1)a^{t_3} \pmod m, \\ b^2-1 & \equiv y+(b^2-1)a^{t_5} \pmod m. \end{cases}$$

We now discuss the possible matchings at levels $s = 1$ and $s = 2$ in these two cases.

Case 1 In this case, the support matrix becomes

$$\begin{bmatrix} 0 & 0 & x & x \\ b-1 & (b-1)a^{t_2} & b-1 & (b-1)+\frac{b}{b+1}x \\ b^2-1 & (b^2-1)a^{t_2} & b^2-1-bx & b^2-1 \\ y & y & z & z \\ y+(b-1)a^{t_5} & y+(b-1)a^{t_6} & z+(b-1)a^{t_7} & z+(b-1)a^{t_8} \\ y+(b^2-1)a^{t_5} & y+(b^2-1)a^{t_6} & z+(b^2-1)a^{t_7} & z+(b^2-1)a^{t_8} \end{bmatrix}.$$

Since $j \geq 3$, we have $b+1 \not\equiv 0 \pmod m$, so the above expressions are well defined.

At level $s = 1$, up to relabeling, there are two possible matchings which contain no pair already matched at level $s = 0$:

$$\text{Case 1-1:} \quad \begin{cases} (b-1) + \dfrac{b}{b+1}x \equiv (b-1)a^{t_2} \pmod m, \\ y + (b-1)a^{t_5} \equiv z + (b-1)a^{t_7} \pmod m, \\ y + (b-1)a^{t_6} \equiv z + (b-1)a^{t_8} \pmod m. \end{cases}$$

$$\text{Case 1-2:} \quad \begin{cases} (b-1) + \dfrac{b}{b+1}x \equiv z + (b-1)a^{t_7} \pmod m, \\ (b-1)a^{t_2} \equiv y + (b-1)a^{t_5} \pmod m, \\ y + (b-1)a^{t_6} \equiv z + (b-1)a^{t_8} \pmod m. \end{cases}$$

Case 1-1 The support matrix becomes

$$\begin{bmatrix} 0 & 0 & x & x & y \\ b-1 & b-1+\frac{b}{b+1}x & b-1 & b-1+\frac{b}{b+1}x & y+(b-1)a^{t_5} \\ b^2-1 & b^2-1+bx & b^2-1-bx & b^2-1 & y+(b^2-1)a^{t_5} \end{bmatrix}$$
$$\begin{matrix} y & z & z \\ y+(b-1)a^{t_6} & y+(b-1)a^{t_5} & y+(b-1)a^{t_6} \\ y+(b^2-1)a^{t_6} & (b+1)y-bz+(b^2-1)a^{t_5} & (b+1)y-bz+(b^2-1)a^{t_6} \end{matrix}\Bigg].$$

At level $s = 2$, if $b^2 - 1 + bx \equiv b^2 - 1 - bx$, then $x \equiv 0 \pmod m$, which forces $c_1 = c_3$, a contradiction. Hence, up to relabeling, the only remaining subcase is the following.

Case 1-1-1

$$\begin{cases} b^2-1-bx & \equiv y + (b^2-1)a^{t_5} \pmod m, \\ (b+1)y - bz + (b^2-1)a^{t_5} & \equiv y + (b^2-1)a^{t_6} \pmod m, \\ (b+1)y - bz + (b^2-1)a^{t_6} & \equiv b^2 - 1 + bx \pmod m. \end{cases}$$

Adding the three congruences gives $y \equiv x + z \pmod m$. Substituting this into the third congruence gives $y + (b^2-1)a^{t_6} \equiv b^2 - 1 \pmod m$, i.e., $c_6 \sim_2 c_1$. Thus a matching block of size four occurs at level $s = 2$, namely

$$\{c_1, c_4, c_6, c_7\} \in M_2.$$

This gives a legal matching configuration.

Case 1-2 The support matrix becomes

$$\begin{bmatrix} 0 & 0 & x & x & y \\ b-1 & (b-1)a^{t_2} & b-1 & b-1+\frac{b}{b+1}x & (b-1)a^{t_2} \\ b^2-1 & (b^2-1)a^{t_2} & b^2-1-bx & b^2-1 & -by+(b^2-1)a^{t_2} \end{bmatrix}$$
$$\begin{matrix} y & z & z \\ y+(b-1)a^{t_6} & b-1+\frac{b}{b+1}x & y+(b-1)a^{t_6} \\ y+(b^2-1)a^{t_6} & bx-bz+b^2-1 & (b+1)y-bz+(b^2-1)a^{t_6} \end{matrix}\Bigg].$$

At level $s = 2$, up to relabeling, there are six possible subcases.

Case 1-2-1

$$\begin{cases} b^2-1-bx & \equiv (b^2-1)a^{t_2} \pmod m, \\ -by+(b^2-1)a^{t_2} & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m, \\ y+(b^2-1)a^{t_6} & \equiv bx-bz+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $2b(z-y-x) \equiv 0 \pmod m$, and hence $z \equiv x+y \pmod m$. Substituting this into the third congruence gives $y+(b-1)a^{t_6} \equiv b-1 \pmod m$, i.e., $c_6 \sim_1 c_1$. Thus a matching block of size four occurs at level $s=1$:

$$\{c_1, c_3, c_6, c_8\} \in M_1.$$

This gives a legal matching configuration.

Case 1-2-2

$$\begin{cases} y+(b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ b^2-1-bx & \equiv bx-bz+b^2-1 \pmod m, \\ -by+(b^2-1)a^{t_2} & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m. \end{cases}$$

The second congruence gives $z \equiv 2x \pmod m$, and adding the first and the third congruences gives $z \equiv 2y \pmod m$. Hence $x \equiv y \pmod m$. Therefore a matching block of size four occurs at level $s=0$:

$$\{c_3, c_4, c_5, c_6\} \in M_0.$$

This gives a legal matching configuration.

Case 1-2-3

$$\begin{cases} y+(b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ b^2-1-bx & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m, \\ -by+(b^2-1)a^{t_2} & \equiv bx-bz+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $z \equiv x+y \pmod m$. Substituting this into the third congruence gives $a^{t_2} \equiv 1 \pmod m$, and hence $c_1 = c_2$ by Lemma 13 of the main manuscript, a contradiction.

Case 1-2-4

$$\begin{cases} (b^2-1)a^{t_2} & \equiv bx-bz+b^2-1 \pmod m, \\ b^2-1-bx & \equiv y+(b^2-1)a^{t_6} \pmod m, \\ (b+1)y-bz+(b^2-1)a^{t_6} & \equiv -by+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

Adding the three congruences gives $x \equiv y \pmod m$. Substituting this into the second congruence gives $y+(b-1)a^{t_6} \equiv b-1 \pmod m$, i.e. $c_6 \sim_1 c_1$. Since $c_1 \sim_1 c_3$ and $c_3 \sim_2 c_6$, we obtain $c_3 = c_6$ from Lemma 14 of the main manuscript, a contradiction.

Case 1-2-5

$$\begin{cases} (b+1)y-bz+(b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ -by+(b^2-1)a^{t_2} & \equiv b^2-1-bx \pmod m, \\ bx-bz+b^2-1 & \equiv y+(b^2-1)a^{t_6} \pmod m. \end{cases}$$

Adding the three congruences gives $x \equiv z \pmod m$, i.e., $\{c_3, c_4, c_7, c_8\} \in M_0$, which forces $c_4 = c_7$ from $c_4 \sim_0 c_7$ and $c_4 \sim_1 c_7$ by Lemma 14 of the main manuscript, a contradiction.

Case 1-2-6

$$\begin{cases} (b+1)y - bz + (b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ -by + (b^2-1)a^{t_2} & \equiv bx - bz + b^2 - 1 \pmod m, \\ b^2 - 1 - bx & \equiv y + (b^2-1)a^{t_6} \pmod m. \end{cases}$$

Adding the three congruences gives $x \equiv 0 \pmod m$, i.e., $\{c_1, c_2, c_3, c_4\} \in M_0$ which forces $c_1 = c_3$ from $c_1 \sim_0 c_3$ and $c_1 \sim_1 c_3$ by Lemma 14 of the main manuscript, a contradiction.

Case 2 In this case, the support matrix becomes

$$\left[\begin{array}{cccc} 0 & 0 & x & x \\ b-1 & (b-1)a^{t_2} & b-1 & x+(b-1)a^{t_4} \\ b^2-1 & (b^2-1)a^{t_2} & b^2-1-bx & x+(b^2-1)a^{t_4} \end{array}\right.$$
$$\left.\begin{array}{cccc} y & y & z & z \\ b-1+\frac{b}{b+1}y & y+(b-1)a^{t_6} & z+(b-1)a^{t_7} & z+(b-1)a^{t_8} \\ b^2-1 & y+(b^2-1)a^{t_6} & z+(b^2-1)a^{t_7} & z+(b^2-1)a^{t_8} \end{array}\right].$$

At level $s = 1$, up to relabeling, there are five possible matchings which contain no pair already matched at level $s = 0$:

$$\text{Case 2-1:} \quad \begin{cases} (b-1)a^{t_2} \equiv x + (b-1)a^{t_4} \pmod m, \\ b - 1 + \dfrac{b}{b+1}y \equiv z + (b-1)a^{t_7} \pmod m, \\ y + (b-1)a^{t_6} \equiv z + (b-1)a^{t_8} \pmod m. \end{cases}$$

$$\text{Case 2-2:} \quad \begin{cases} (b-1)a^{t_2} \equiv b - 1 + \dfrac{b}{b+1}y \pmod m, \\ x + (b-1)a^{t_4} \equiv z + (b-1)a^{t_7} \pmod m, \\ y + (b-1)a^{t_6} \equiv z + (b-1)a^{t_8} \pmod m. \end{cases}$$

$$\text{Case 2-3:} \quad \begin{cases} (b-1)a^{t_2} \equiv y + (b-1)a^{t_6} \pmod m, \\ x + (b-1)a^{t_4} \equiv z + (b-1)a^{t_7} \pmod m, \\ b - 1 + \dfrac{b}{b+1}y \equiv z + (b-1)a^{t_8} \pmod m. \end{cases}$$

$$\text{Case 2-4:} \quad \begin{cases} (b-1)a^{t_2} \equiv z + (b-1)a^{t_7} \pmod m, \\ x + (b-1)a^{t_4} \equiv b - 1 + \dfrac{b}{b+1}y \pmod m, \\ y + (b-1)a^{t_6} \equiv z + (b-1)a^{t_8} \pmod m. \end{cases}$$

$$\text{Case 2-5:} \quad \begin{cases} (b-1)a^{t_2} \equiv z + (b-1)a^{t_7} \pmod m, \\ z + (b-1)a^{t_8} \equiv b - 1 + \dfrac{b}{b+1}y \pmod m, \\ x + (b-1)a^{t_4} \equiv y + (b-1)a^{t_6} \pmod m. \end{cases}$$

Case 2-1 The support matrix becomes

$$\left[\begin{array}{cccc} 0 & 0 & x & x \\ b-1 & (b-1)a^{t_2} & b-1 & (b-1)a^{t_2} \\ b^2-1 & (b^2-1)a^{t_2} & b^2-1-bx & -bx+(b^2-1)a^{t_2} \end{array}\right.$$
$$\left.\begin{array}{cccc} y & y & z & z \\ b-1+\frac{b}{b+1}y & y+(b-1)a^{t_6} & b-1+\frac{b}{b+1}y & y+(b-1)a^{t_6} \\ b^2-1 & y+(b^2-1)a^{t_6} & by-bz+b^2-1 & (b+1)y-bz+(b^2-1)a^{t_6} \end{array}\right].$$

At level $s = 2$, up to relabeling, there are seven possible subcases.

Case 2-1-1

$$\begin{cases} b^2-1-bx & \equiv (b^2-1)a^{t_2} \pmod m, \\ -bx+(b^2-1)a^{t_2} & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m, \\ y+(b^2-1)a^{t_6} & \equiv by-bz+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $z \equiv x+y \pmod m$. Substituting this into the first and third congruences gives $y+(b^2-1)a^{t_6} \equiv (b^2-1)a^{t_2} \pmod m$, i.e., $c_6 \sim_2 c_2$. Therefore a matching block of size four occurs at level $s=2$:

$$\{c_2, c_3, c_6, c_7\} \in M_2.$$

This gives a legal matching configuration.

Case 2-1-2

$$\begin{cases} b^2-1-bx & \equiv y+(b^2-1)a^{t_6} \pmod m, \\ (b^2-1)a^{t_2} & \equiv by-bz+b^2-1 \pmod m, \\ (b+1)y-bz+(b^2-1)a^{t_6} & \equiv -bx+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

This case does not force repeated support columns, and hence it gives a legal matching configuration.

Case 2-1-3

$$\begin{cases} b^2-1-bx & \equiv y+(b^2-1)a^{t_6} \pmod m, \\ (b+1)y-bz+(b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ -bx+(b^2-1)a^{t_2} & \equiv by-bz+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $-2bx \equiv 0 \pmod m$, and hence $x \equiv 0 \pmod m$. Thus $c_1 = c_3$ from $c_1 \sim_0 c_3$ and $c_1 \sim_1 c_3$ by Lemma 14 of the main manuscript, a contradiction.

Case 2-1-4

$$\begin{cases} b^2-1-bx & \equiv by-bz+b^2-1 \pmod m, \\ y+(b^2-1)a^{t_6} & \equiv -bx+(b^2-1)a^{t_2} \pmod m, \\ (b^2-1)a^{t_2} & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(y-z) \pmod m$, and hence $z \equiv y \pmod m$. Thus $c_6 = c_8$ from $c_6 \sim_0 c_8$ and $c_6 \sim_1 c_8$ by Lemma 14 of the main manuscript, a contradiction.

Case 2-1-5

$$\begin{cases} b^2-1-bx & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m, \\ (b^2-1)a^{t_2} & \equiv by-bz+b^2-1 \pmod m, \\ y+(b^2-1)a^{t_6} & \equiv -bx+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(y-z) \pmod m$, and hence $z \equiv y$. Similarly, $c_6 = c_8$, a contradiction.

Case 2-1-6

$$\begin{cases} b^2-1-bx & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m, \\ y+(b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ -bx+(b^2-1)a^{t_2} & \equiv by-bz+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $-2bx \equiv 2b(y-z) \pmod m$, and hence $z \equiv x+y \pmod m$. Substituting this into the first congruence gives $y+(b^2-1)a^{t_6} \equiv b^2-1$, i.e., $c_6 \sim_2 c_1$. Since $c_1 \sim_2 c_5$ and $c_5 \sim_0 c_6$, we obtain $c_5 = c_6$ by Lemma 14 of the main manuscript, a contradiction.

Case 2-1-7

$$\begin{cases} -bx + (b^2-1)a^{t_2} & \equiv (b+1)y - bz + (b^2-1)a^{t_6} \pmod m, \\ y + (b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ b^2 - 1 - bx & \equiv by - bz + b^2 - 1 \pmod m. \end{cases}$$

Adding the three congruences gives $-2bx \equiv 2b(y-z) \pmod m$, and hence $z \equiv x+y \pmod m$. This case does not force repeated support columns, and therefore it gives a legal matching configuration.

Case 2-2 The support matrix becomes

$$\left[\begin{array}{ccccc} 0 & 0 & x & x & y \\ b-1 & b-1+\frac{b}{b+1}y & b-1 & x+(b-1)a^{t_4} & b-1+\frac{b}{b+1}y \\ b^2-1 & by+b^2-1 & b^2-1-bx & x+(b^2-1)a^{t_4} & b^2-1 \end{array}\right.$$

$$\left.\begin{array}{ccc} y & z & z \\ y+(b-1)a^{t_6} & x+(b-1)a^{t_4} & y+(b-1)a^{t_6} \\ y+(b^2-1)a^{t_6} & (b+1)x-bz+(b^2-1)a^{t_4} & (b+1)y-bz+(b^2-1)a^{t_6} \end{array}\right].$$

At level $s=2$, up to relabeling, there are six possible subcases.

Case 2-2-1

$$\begin{cases} b^2-1-bx & \equiv by + b^2 - 1 \pmod m, \\ x + (b^2-1)a^{t_4} & \equiv (b+1)y - bz + (b^2-1)a^{t_6} \pmod m, \\ y + (b^2-1)a^{t_6} & \equiv (b+1)x - bz + (b^2-1)a^{t_4} \pmod m. \end{cases}$$

The first congruence gives $x+y \equiv 0 \pmod m$, while adding the three congruences gives $z \equiv x+y \pmod m$. Hence $z \equiv 0 \pmod m$. Therefore a matching block of size four occurs at level $s=0$:

$$\{c_1, c_2, c_7, c_8\} \in M_0.$$

This gives a legal matching configuration.

Case 2-2-2

$$\begin{cases} b^2-1-bx & \equiv y + (b^2-1)a^{t_6} \pmod m, \\ (b+1)y - bz + (b^2-1)a^{t_6} & \equiv x + (b^2-1)a^{t_4} \pmod m, \\ (b+1)x - bz + (b^2-1)a^{t_4} & \equiv by + b^2 - 1 \pmod m. \end{cases}$$

Adding the three congruences gives $-2bz \equiv 0$, and hence $z \equiv 0 \pmod m$. Then the pair $\{c_2, c_7\}$ is matched at $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_2 = c_7$, a contradiction.

Case 2-2-3

$$\begin{cases} b^2-1-bx & \equiv (b+1)x - bz + (b^2-1)a^{t_4} \pmod m, \\ y + (b^2-1)a^{t_6} & \equiv by + (b^2-1) \pmod m, \\ x + (b^2-1)a^{t_4} & \equiv (b+1)y - bz + (b^2-1)a^{t_6} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x+y-z) \pmod m$, and hence $z \equiv x+y \pmod m$. Substituting this into the first congruence and dividing by $b+1$ gives

$$b - 1 + \frac{b}{b+1}y \equiv x + (b-1)a^{t_4} \pmod m.$$

Therefore a matching block of size four occurs at level $s=1$:

$$\{c_2, c_4, c_5, c_7\} \in M_1.$$

This gives a legal matching configuration.

Case 2-2-4

$$\begin{cases} b^2-1-bx & \equiv (b+1)x - bz + (b^2-1)a^{t_4} \pmod m, \\ x + (b^2-1)a^{t_4} & \equiv y + (b^2-1)a^{t_6} \pmod m, \\ (b+1)y - bz + (b^2-1)a^{t_6} & \equiv by + (b^2-1) \pmod m. \end{cases}$$

Adding the three congruences gives $-2bx \equiv 0 \pmod m$, and hence $x \equiv 0 \pmod m$. Then the pair $\{c_1, c_3\}$ is matched at levels $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_1 = c_3$, a contradiction.

Case 2-2-5

$$\begin{cases} b^2-1-bx & \equiv (b+1)y - bz + (b^2-1)a^{t_6} \pmod m, \\ x + (b^2-1)a^{t_4} & \equiv by + (b^2-1) \pmod m, \\ y + (b^2-1)a^{t_6} & \equiv (b+1)x - bz + (b^2-1)a^{t_4} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x+y-z) \pmod m$, and hence $z \equiv x+y \pmod m$. Substituting this into the first congruence gives

$$b^2 - 1 \equiv y + (b^2-1)a^{t_6} \pmod m.$$

Then the pair $\{c_5, c_6\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_5 = c_6$, a contradiction.

Case 2-2-6

$$\begin{cases} b^2-1-bx & \equiv (b+1)y - bz + (b^2-1)a^{t_6} \pmod m, \\ y + (b^2-1)a^{t_6} & \equiv x + (b^2-1)a^{t_4} \pmod m, \\ (b+1)x - bz + (b^2-1)a^{t_4} & \equiv by + b^2 - 1 \pmod m. \end{cases}$$

Adding the three congruences gives $2by \equiv 0$, and hence $y \equiv 0 \pmod m$. Then the pair $\{c_2, c_5\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_2 = c_5$, a contradiction.

Case 2-3 The support matrix becomes:

$$\begin{bmatrix} 0 & 0 & x & x & y \\ b-1 & (b-1)a^{t_2} & b-1 & x+(b-1)a^{t_4} & b-1+\frac{b}{b+1}y \\ b^2-1 & (b^2-1)a^{t_2} & b^2-1-bx & x+(b^2-1)a^{t_4} & b^2-1 \\ y & z & z \\ (b-1)a^{t_2} & x+(b-1)a^{t_4} & b-1+\frac{b}{b+1}y \\ -by+(b^2-1)a^{t_2} & (b+1)x-bz+(b^2-1)a^{t_4} & by-bz+b^2-1 \end{bmatrix}.$$

At level $s = 2$, up to relabeling, there are six possible subcases.

Case 2-3-1

$$\begin{cases} b^2-1-bx & \equiv (b^2-1)a^{t_2} \pmod m, \\ x + (b^2-1)a^{t_4} & \equiv by - bz + b^2 - 1 \pmod m, \\ -by + (b^2-1)a^{t_2} & \equiv (b+1)x - bz + (b^2-1)a^{t_4} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x+y-z) \pmod m$, and hence $z \equiv x+y \pmod m$. Substituting this into the first congruence gives

$$b^2 - 1 - bx \equiv by - bz + b^2 - 1 \equiv x + (b^2-1)a^{t_4} \pmod m.$$

Then the pair $\{c_3, c_4\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_3 = c_4$, a contradiction.

Case 2-3-2

$$\begin{cases} b^2-1-bx & \equiv -by+(b^2-1)a^{t_2} \pmod m, \\ x+(b^2-1)a^{t_4} & \equiv by-bz+b^2-1 \pmod m, \\ (b^2-1)a^{t_2} & \equiv (b+1)x-bz+(b^2-1)a^{t_4} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x-z) \pmod m$, and hence $x \equiv z \pmod m$. Then the pair $\{c_4, c_7\}$ is matched at levels $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_4 = c_7$, a contradiction.

Case 2-3-3

$$\begin{cases} b^2-1-bx & \equiv (b+1)x-bz+(b^2-1)a^{t_4} \pmod m, \\ x+(b^2-1)a^{t_4} & \equiv (b^2-1)a^{t_2} \pmod m, \\ -by+(b^2-1)a^{t_2} & \equiv by-bz+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x+y-z) \pmod m$, and hence $z \equiv x+y \pmod m$. Moreover, the first congruence gives

$$(b+1)x-bz+(b^2-1)a^{t_4} \equiv b^2-1-bx \equiv by-bz+b^2-1 \pmod m,$$

i.e. $c_7 \sim_2 c_8$. Because $c_7 \sim_0 c_8$ and $c_7 \sim_2 c_8$, this forces $c_7 = c_8$ by Lemma 14 of the main manuscript, a contradiction.

Case 2-3-4

$$\begin{cases} b^2-1-bx & \equiv (b+1)x-bz+(b^2-1)a^{t_4} \pmod m, \\ (b^2-1)a^{t_2} & \equiv by-bz+b^2-1 \pmod m, \\ x+(b^2-1)a^{t_4} & \equiv -by+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x-z) \pmod m$, and hence $x \equiv z \pmod m$. Then the pair $\{c_4, c_7\}$ is matched at levels $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_4 = c_7$, a contradiction.

Case 2-3-5

$$\begin{cases} b^2-1-bx & \equiv by-bz+b^2-1 \pmod m, \\ (b^2-1)a^{t_2} & \equiv x+(b^2-1)a^{t_4} \pmod m, \\ (b+1)x-bz+(b^2-1)a^{t_4} & \equiv -by+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

The first congruence gives $z \equiv x+y \pmod m$. This case does not force repeated support columns, and therefore it gives a legal matching configuration.

Case 2-3-6

$$\begin{cases} b^2-1-bx & \equiv by-bz+b^2-1 \pmod m, \\ (b^2-1)a^{t_2} & \equiv (b+1)x-bz+(b^2-1)a^{t_4} \pmod m, \\ x+(b^2-1)a^{t_4} & \equiv -by+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

The first congruence gives $z \equiv x+y \pmod m$. Adding the three congruences gives $0 \equiv 2b(x-z) \pmod m$, and hence $x \equiv z \pmod m$. Therefore $y \equiv 0 \pmod m$, i.e., $c_1 \sim_0 c_5$. Then the pair $\{c_1, c_5\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_1 = c_5$, a contradiction.

Case 2-4 The support matrix becomes

$$\begin{bmatrix} 0 & 0 & x & x & y \\ b-1 & (b-1)a^{t_2} & b-1 & b-1+\frac{b}{b+1}y & b-1+\frac{b}{b+1}y \\ b^2-1 & (b^2-1)a^{t_2} & b^2-1-bx & by-bx+b^2-1 & b^2-1 \\ y & z & z & & \\ y+(b-1)a^{t_6} & (b-1)a^{t_2} & y+(b-1)a^{t_6} & & \\ y+(b^2-1)a^{t_6} & -bz+(b^2-1)a^{t_2} & (b+1)y-bz+(b^2-1)a^{t_6} & & \end{bmatrix}.$$

At level $s = 2$, up to relabeling, there are six possible subcases.

Case 2-4-1

$$\begin{cases} b^2-1-bx & \equiv (b^2-1)a^{t_2} \pmod m, \\ -bz+(b^2-1)a^{t_2} & \equiv y+(b^2-1)a^{t_6} \pmod m, \\ (b+1)y-bz+(b^2-1)a^{t_6} & \equiv by-bx+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $-2bz \equiv 0 \pmod m$, and hence $z \equiv 0 \pmod m$, i.e., $c_2 \sim_0 c_7$. Then the pair $\{c_2, c_7\}$ is matched at two distinct levels $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_2 = c_7$, a contradiction.

Case 2-4-2

$$\begin{cases} y+(b^2-1)a^{t_6} & \equiv b^2-1-bx \pmod m, \\ (b^2-1)a^{t_2} & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m, \\ by-bx+b^2-1 & \equiv -bz+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv -2bz \pmod m$, and hence $z \equiv 0 \pmod m$. Similarly, this forces $c_2 = c_7$, a contradiction.

Case 2-4-3

$$\begin{cases} b^2-1-bx & \equiv -bz+(b^2-1)a^{t_2} \pmod m, \\ (b^2-1)a^{t_2} & \equiv y+(b^2-1)a^{t_6} \pmod m, \\ (b+1)y-bz+(b^2-1)a^{t_6} & \equiv by-bx+b^2-1 \pmod m. \end{cases}$$

This case does not force repeated support columns, and therefore it gives a legal matching configuration.

Case 2-4-4

$$\begin{cases} b^2-1-bx & \equiv -bz+(b^2-1)a^{t_2} \pmod m, \\ y+(b^2-1)a^{t_6} & \equiv by-bx+b^2-1 \pmod m, \\ (b^2-1)a^{t_2} & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(y-z) \pmod m$, and hence $y \equiv z \pmod m$, i.e., $c_6 \sim_0 c_8$. Then the pair $\{c_6, c_8\}$ is matched at $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_6 = c_8$, a contradiction.

Case 2-4-5

$$\begin{cases} b^2-1-bx & \equiv (b+1)y-bz+(b^2-1)a^{t_6} \pmod m, \\ (b^2-1)a^{t_2} & \equiv by-bx+b^2-1 \pmod m, \\ y+(b^2-1)a^{t_6} & \equiv -bz+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(y-z) \pmod m$, and hence $y \equiv z \pmod m$. Similarly, by Lemma 14 of the main manuscript, this forces $c_6 = c_8$, a contradiction.

Case 2-4-6

$$\begin{cases} b^2-1-bx & \equiv (b+1)y - bz + (b^2-1)a^{t_6} \pmod m, \\ y + (b^2-1)a^{t_6} & \equiv (b^2-1)a^{t_2} \pmod m, \\ -bz + (b^2-1)a^{t_2} & \equiv by - bx + b^2 - 1 \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2by \pmod m$, and hence $y \equiv 0 \pmod m$. Then the pair $\{c_1, c_5\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_1 = c_5$, a contradiction.

Case 2-5 The support matrix becomes

$$\left[\begin{array}{cccc} 0 & 0 & x & x \\ b-1 & (b-1)a^{t_2} & b-1 & x+(b-1)a^{t_4} \\ b^2-1 & (b^2-1)a^{t_2} & b^2-1-bx & x+(b^2-1)a^{t_4} \end{array}\right.$$
$$\left.\begin{array}{cccc} y & y & z & z \\ b-1+\frac{b}{b+1}y & x+(b-1)a^{t_4} & (b-1)a^{t_2} & b-1+\frac{b}{b+1}y \\ b^2-1 & (b+1)x-by+(b^2-1)a^{t_4} & -bz+(b^2-1)a^{t_2} & by-bz+b^2-1 \end{array}\right].$$

At level $s = 2$, up to relabeling, there are seven possible subcases.

Case 2-5-1

$$\begin{cases} b^2-1-bx & \equiv (b^2-1)a^{t_2} \pmod m, \\ -bz + (b^2-1)a^{t_2} & \equiv x + (b^2-1)a^{t_4} \pmod m, \\ (b+1)x - by + (b^2-1)a^{t_4} & \equiv by - bz + b^2 - 1 \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2by \pmod m$, and hence $y \equiv 0 \pmod m$. Then the pair $\{c_1, c_5\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_1 = c_5$, a contradiction.

Case 2-5-2

$$\begin{cases} b^2-1-bx & \equiv (b^2-1)a^{t_2} \pmod m, \\ x + (b^2-1)a^{t_4} & \equiv by - bz + b^2 - 1 \pmod m, \\ -bz + (b^2-1)a^{t_2} & \equiv (b+1)x - by + (b^2-1)a^{t_4} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2bx \pmod m$, and hence $x \equiv 0 \pmod m$. Then the pair $\{c_1, c_3\}$ is matched at levels $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_1 = c_3$, a contradiction.

Case 2-5-3

$$\begin{cases} b^2-1-bx & \equiv (b+1)x - by + (b^2-1)a^{t_4} \pmod m, \\ (b^2-1)a^{t_2} & \equiv by - bz + b^2 - 1 \pmod m, \\ x + (b^2-1)a^{t_4} & \equiv -bz + (b^2-1)a^{t_2} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x - z) \pmod m$, and hence $x \equiv z \pmod m$. Then the pair $\{c_4, c_7\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_4 = c_7$, a contradiction.

Case 2-5-4

$$\begin{cases} b^2-1-bx & \equiv -bz+(b^2-1)a^{t_2} \pmod m, \\ (b^2-1)a^{t_2} & \equiv x+(b^2-1)a^{t_4} \pmod m, \\ (b+1)x-by+(b^2-1)a^{t_4} & \equiv by-bz+b^2-1 \pmod m. \end{cases}$$

Adding the three congruences gives $2by \equiv 2bz \pmod m$, and hence $y \equiv z \pmod m$. Then the pair $\{c_5, c_8\}$ is matched at levels $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_5 = c_8$, a contradiction.

Case 2-5-5

$$\begin{cases} b^2-1-bx & \equiv -bz+(b^2-1)a^{t_2} \pmod m, \\ x+(b^2-1)a^{t_4} & \equiv by-bz+b^2-1 \pmod m, \\ (b^2-1)a^{t_2} & \equiv (b+1)x-by+(b^2-1)a^{t_4} \pmod m. \end{cases}$$

Adding the three congruences gives $0 \equiv 2b(x-z) \pmod m$, and hence $x \equiv z \pmod m$. Substituting this into the first congruence gives

$$b^2-1 \equiv (b^2-1)a^{t_2} \pmod m,$$

i.e., $c_1 \sim_2 c_2$. Then the pair $\{c_1, c_2\}$ is matched at $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_1 = c_2$, a contradiction.

Case 2-5-6

$$\begin{cases} b^2-1-bx & \equiv by-bz+b^2-1 \pmod m, \\ (b^2-1)a^{t_2} & \equiv x+(b^2-1)a^{t_4} \pmod m, \\ (b+1)x-by+(b^2-1)a^{t_4} & \equiv -bz+(b^2-1)a^{t_2} \pmod m. \end{cases}$$

The first congruence gives $z \equiv x+y \pmod m$. Adding the three congruences gives $0 \equiv 2b(y-z)$, and hence $y \equiv z \pmod m$. Therefore $x \equiv 0 \pmod m$. Then the pair $\{c_1, c_3\}$ is matched at levels $s = 0, 1$. By Lemma 14 of the main manuscript, this forces $c_1 = c_3$, a contradiction.

Case 2-5-7

$$\begin{cases} b^2-1-bx & \equiv by-bz+b^2-1 \pmod m, \\ x+(b^2-1)a^{t_4} & \equiv -bz+(b^2-1)a^{t_2} \pmod m, \\ (b^2-1)a^{t_2} & \equiv (b+1)x-by+(b^2-1)a^{t_4} \pmod m. \end{cases}$$

The first congruence gives $z \equiv x+y \pmod m$. Adding the three congruences gives $0 \equiv 2b(x-z) \pmod m$, and hence $x \equiv z \pmod m$. Therefore $y \equiv 0 \pmod m$. Then the pair $\{c_1, c_5\}$ is matched at levels $s = 0, 2$. By Lemma 14 of the main manuscript, this forces $c_1 = c_5$, a contradiction.

The enumeration above contains 39 nontrivial subcases. Every discarded subcase forces either repeated support columns or a pair of distinct support columns matched at two different levels. The remaining subcases give the following ten legal matching configurations:

1-1-1, 1-2-1, 1-2-2, 2-1-1, 2-1-2, 2-1-7, 2-2-1, 2-2-3, 2-3-5, 2-4-3.

The forced 4-matchings have been retained throughout the enumeration. Together with the pairwise-refinement procedure, this accounts for both block-size types in Lemma 20 of the main manuscript.

Among the ten matching configurations, consider the permutations

$$\pi_1 = (1\,2)(3\,4)(5\,8)(6\,7),$$

$$\pi_2 = (1\,2)(3\,5)(4\,6)(7\,8),$$
$$\pi_3 = (1\,3)(2\,4)(5\,7)(6\,8).$$

A direct verification shows that π_1 is a weight-preserving graph isomorphism between the matching graphs of Cases 1-1-1 and 2-1-1, π_2 is a weight-preserving graph isomorphism between those of Cases 1-2-1 and 2-2-3, and π_3 is a weight-preserving graph isomorphism between those of Cases 1-2-2 and 2-2-1.

By Lemma 19 of the main manuscript, each of these pairs describes the same support configurations up to a weight-preserving relabeling.

Consequently, it is sufficient to retain the seven representatives

$$\text{1-1-1}, \quad \text{1-2-1}, \quad \text{1-2-2}, \quad \text{2-1-2}, \quad \text{2-1-7}, \quad \text{2-3-5}, \quad \text{2-4-3}.$$

We denote the above seven cases by Case I–Case VII, respectively. This proves Proposition 29 of the main manuscript. □

4 Conclusion

The preceding case analysis exhausts the pairwise refinements and all admissible same-level mergings prescribed by the enumeration procedure. Every discarded branch forces repeated support columns, whereas the surviving branches reduce, under weight-preserving relabeling, to exactly the seven representatives listed in Proposition 29. This completes the classification of weight-8 matching configurations.